

Proof by induction - sequences

Basic skills

$$Q1. a_{n+1} = 3a_n - 1 \quad a_1 = 1$$

$$a) a_2 = 3a_1 - 1 = 3 - 1 = 2$$

$$b) a_3 = 3a_2 - 1 = 6 - 1 = 5$$

$$c) a_4 = 3a_3 - 1 = 15 - 1 = 14$$

$$d) a_5 = 42 - 1 = 41$$

$$Q2. a_{n+1} = a_n^2 - a_n - 1 \quad a_1 = 3$$

$$a) a_2 = a_1^2 - a_1 - 1 = 3^2 - 3 - 1 = 5$$

$$b) a_3 = a_2^2 - a_2 - 1 = 5^2 - 5 - 1 = 19$$

$$c) a_4 = a_3^2 - a_3 - 1 = 19^2 - 19 - 1 = 341$$

$$Q3. a_{n+1} = 2a_n - 1 \quad a_k = 1$$

$$a_{k+1} = 2a_k - 1 = 2 - 1 = 1$$

Q4. $a_{n+1} = a_n(a_n - 2)$ $a_k = 4$

$$a_{k+1} = a_k(a_k - 2) \\ = 4 \times (4 - 2) = 8$$

Q5. $a_{n+1} = 3a_n$ $a_k = 3^k$

$$a_{k+1} = 3a_k = 3 \times 3^k = 3^{k+1}$$

Competency skills

recurrence relation

Q6. $a_1 = 1$ $a_{n+1} = a_n + 5$

Formula: $a_n = 5n - 4$ for all n

a) Base case:

$$a_1 = 1, \quad a_1 = 5 \times 1 - 4 = 1$$

So holds

b) $\rightarrow$ use recurrence first

$$a_{k+1} = a_k + 5 = 5k - 4 + 5$$

$\rightarrow$ I.H assumption

$$= 5k+1$$

$$= 5(k+1) - 4$$

So Formula holds

C) As Formula holds for $n=1$
and if true for k then true for
 $k+1$ so true for all n by
mathematical induction

Q7. $a_1 = 2$ $a_{n+1} = 2a_n + 5$

Formula: $a_n = 7 \times 2^{n-1} - 5$

Base case: $a_1 = 2$, $a_1 = 7 \times 2^0 - 5 = 2$
so holds

I.H : Assume formula holds for
some k , Show holds for $k+1$

I.S : $a_{k+1} \overset{\text{recurrence}}{=} 2a_k + 5$

$$= 2(7 \times 2^{k-1} - 5) + 5$$

$$= 7 \times 2 \times 2^{k-1} - 10 + 5$$

$$= 7 \times 2^k - 5$$

as required

Conclusion: As true for $n=1$ and if true for k then true for $k+1$ we have it true for all n by induction

Q8. $a_1 = \frac{3}{2}$ $a_{n+1} = \frac{a_n^2 + 2}{3}$

Statement: $1 < a_n < 2$

Base case: $1 < \frac{3}{2} < 2$ holds

I.H: Assume $1 < a_k < 2$ for some k

I.S: $a_{k+1} = \frac{a_k^2 + 2}{3}$

as $1 < a_k < 2$ by I.H

$$1 < a_k^2 < 4$$

$$\text{So } 3 < a_k^2 + 2 < 6$$

$$\text{So } 1 < \frac{a_k^2 + 2}{3} < 2$$

$$\text{So } 1 < a_{k+1} < 2$$

Conclusion: AS Statement holds for $n=1$ and if holds for k then hold for $k+1$, so true for all n

Advanced skills

Q9. $a_1 = 1, a_{n+1} = (n+1)a_n$

a) $a_{k+1} = (k+1)a_k$

assumption is $a_k > 2^k$

so $(k+1)a_k > (k+1)2^k$

and $(k+1) \geq 2$

as $k \geq 1$

so $(k+1)a_k \geq 2 \cdot 2^k = 2^{k+1}$

so $a_{k+1} > 2^{k+1}$

b) because $a_1 = 1$

so $a_1 \not> 2^1 = 2$

is not greater than

In other words the base case doesn't hold.

Q10. ① ②

a) n objects choose r

Answer is $\binom{n}{r} = {}^nC_r = \frac{n!}{(n-r)!r!}$ let's see why

choices of 1st obj : n
2nd obj : $n-1$
 $\vdots$
 r^{th} obj : $n-r+1$

$$\text{so } n(n-1) \dots (n-r+1) = \frac{n!}{(n-r)!}$$

but these are arranged (order matters)
to remove way of arranging r objects
we $\div r!$

$$\text{so } \frac{n!}{(n-r)! r!}$$

b) Forced to include x

so now out of k balls

choose $r-1$ i.e. $\binom{k}{r-1}$

c) Forced to Not include x
 so now out of k balls
 choose r i.e. $\binom{k}{r}$

d)

All options ← Not including x Including x

$$\binom{k+1}{r} = \binom{k}{r-1} + \binom{k}{r}$$

e) Formula: know

$$\binom{n}{r} = \frac{n!}{(n-r)! r!}$$

$$\binom{1}{1} = 1$$

$$\binom{k+1}{r} = \binom{k}{r-1} + \binom{k}{r}$$

Base case: $(1) = 1$, $\frac{1!}{(1-1)!1!} = \frac{1}{1 \times 1} = 1$
holds

I.H Assume true for $n=k$

I.S

$$\begin{aligned} \binom{k+1}{r} &= \binom{k}{r-1} + \binom{k}{r} \\ &\quad \downarrow \text{I.H} \quad \downarrow \text{I.H} \\ &= \frac{k!}{(k-r+1)! (r-1)!} + \frac{k!}{(k-r)! r!} \\ &= \frac{k! \times r + k! \times (k-r+1)}{(k-r+1)! r!} \\ &= \frac{k! (k+1)}{(k+1-r)! r!} = \frac{(k+1)!}{(k+1-r)! r!} \end{aligned}$$

so by induction formula holds
(normal conclusion)